DevOps



LAB 05

Build, Test, and Package Applications with IntelliJ IDEA and Docker

Submitted To: Sir Waqas Saleem

Submitted By: Ayesha Rehan

Registration No: 2023-BCS-021 Course

Code: CS-649

GitHub Username: ayesha-rehan12

BCS 7-B

Department Of Computer Science

Task 1: Clone and Launch IntelliJ IDEA

```
ayesha-rehan12@ubuntu:~$ cd ~
ayesha-rehan12@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Intellij-Id
a-docker.git
Cloning into 'Intellij-Idea-docker'...
remote: Enumerating objects: 35, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (32/32), done.
remote: Total 35 (delta 16), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (35/35), 15.24 KiB | 15.00 KiB/s, done.
Resolving deltas: 100% (16/16), done.
ayesha-rehan12@ubuntu:~$ cd Intellij-Idea-docker
ayesha-rehan12@ubuntu:~/Intellij-Idea-docker$
```

Figure: Intellij-Idea-docker repository cloned on the Ubuntu host.

```
Browser username: admin (or WEBTOP_USER from your existing .env).
Open https://localhost:3001 in your host computer browser.
ayesha-rehan12@ubuntu:~/Intellij-Idea-docker$ docker compose ps
NAME                            IMAGE                       COMMAND  SERVICE  CREATED         STATUS         PORTS
intellij-idea-docker-intellij-1  intellij-idea-docker-intellij  "/init"  intellij  24 seconds ago  Up 22 seconds  3000/tcp, 127.0.0.1:3001->3001/tcp
ayesha-rehan12@ubuntu:~/Intellij-Idea-docker$
```

Figure: intellij service running with desktop port 3001 (password not shown).

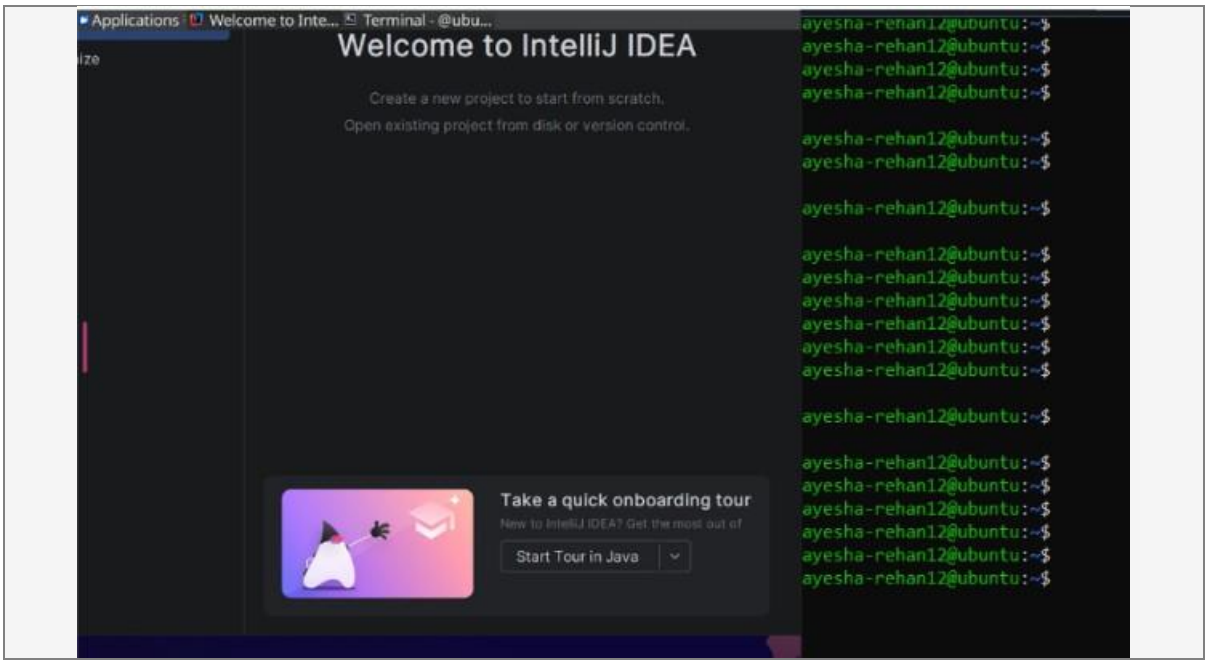


Figure: IntelliJ IDEA open in the browser desktop beside the Ubuntu host terminal.

Task 2: Build and Run the Maven Project

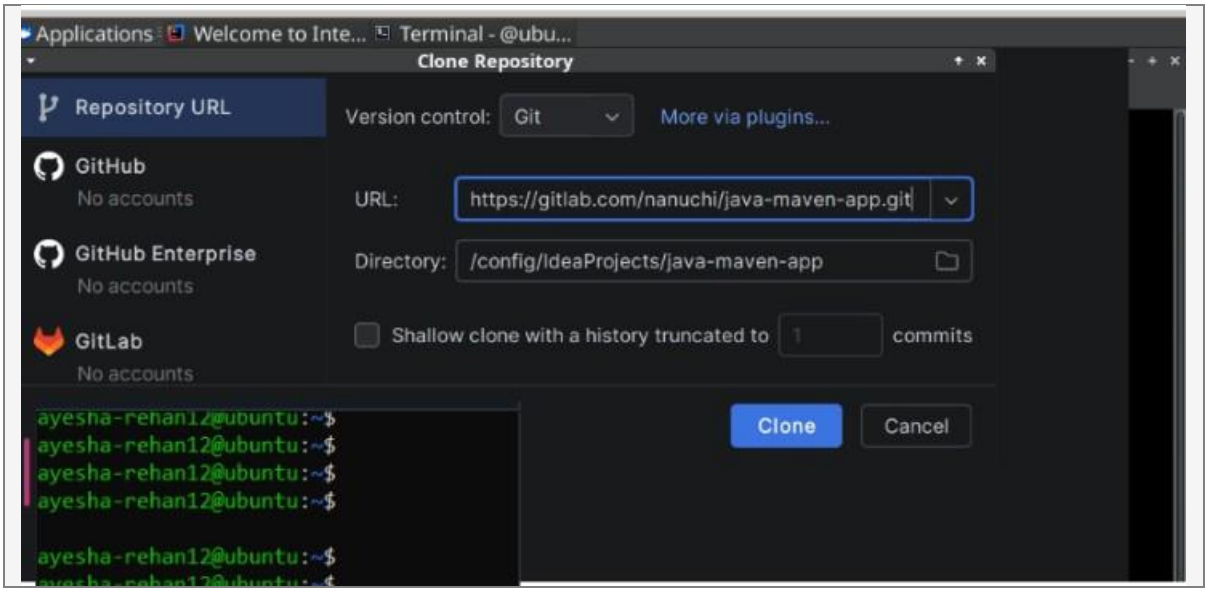


Figure: Clone dialog showing the java-maven-app URL and destination directory.



Figure: Maven installed; version shown in IntelliJ's terminal.

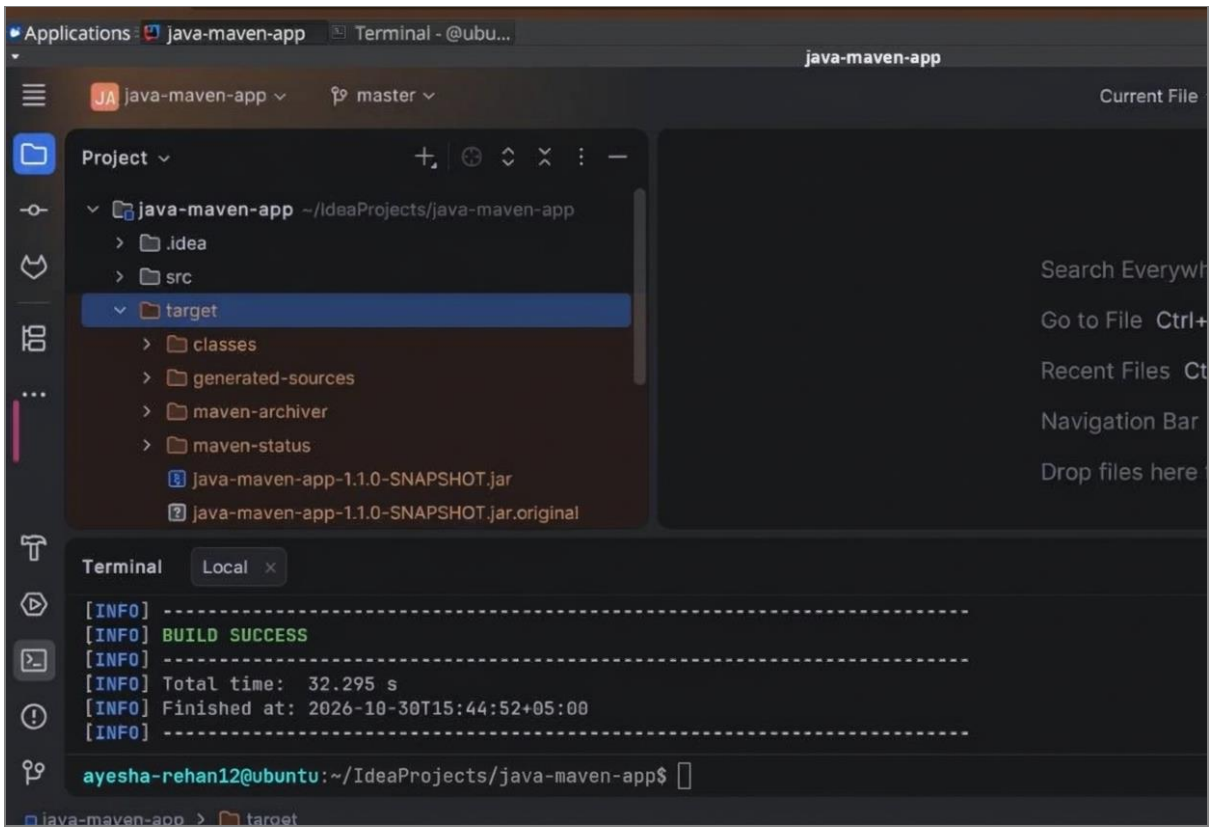


Figure: mvn package completed with BUILD SUCCESS.

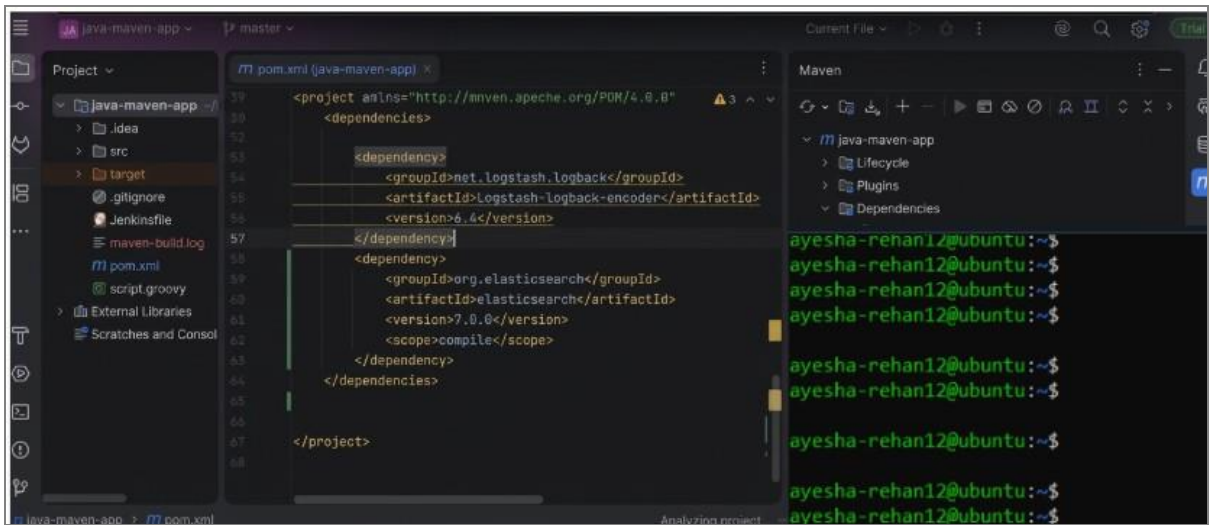


Figure: Elasticsearch 7.0.0 dependency added in pom.xml and synchronized.

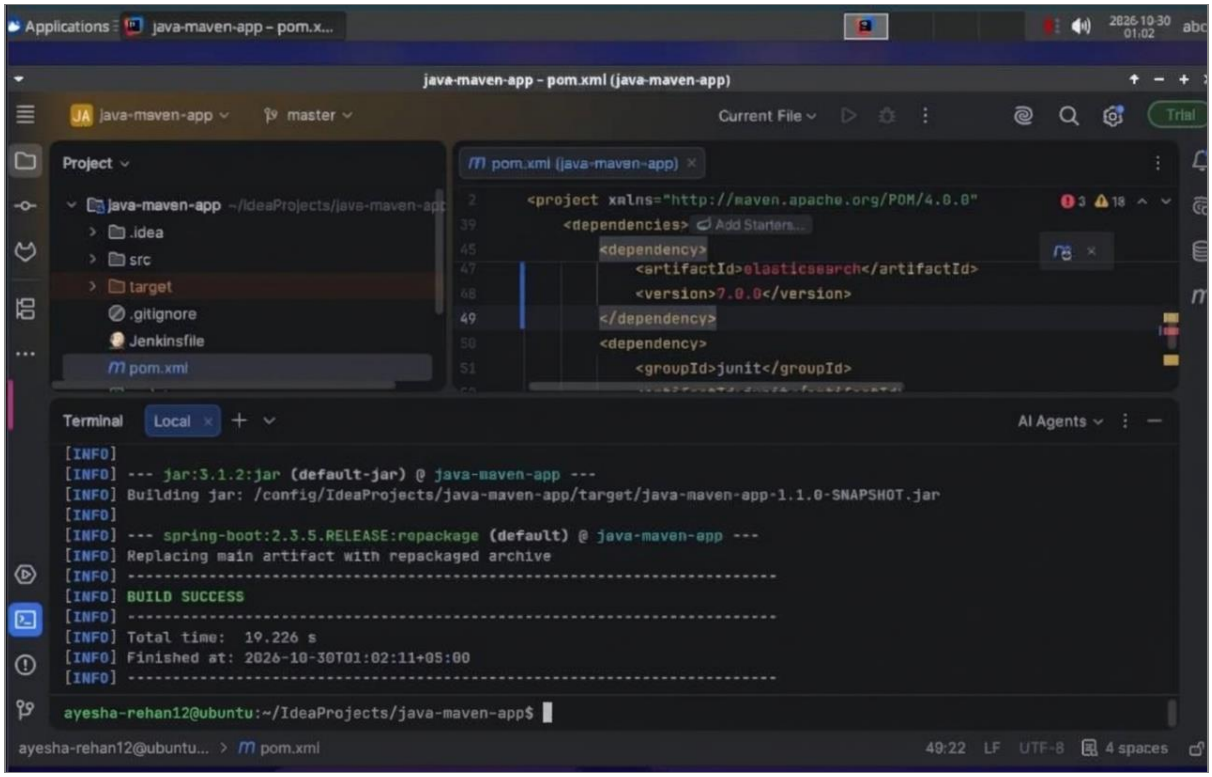


Figure: mvn package re-run after adding the dependency, with BUILD SUCCESS.

```
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  32.295 s
[INFO] Finished at: 2026-10-30T15:44:52+05:00
[INFO] ------------------------------------------------------------------------
ayesha-rehan12@ubuntu:~/IdeaProjects/java-maven-app$
```

Figure: mvn test output and build result.

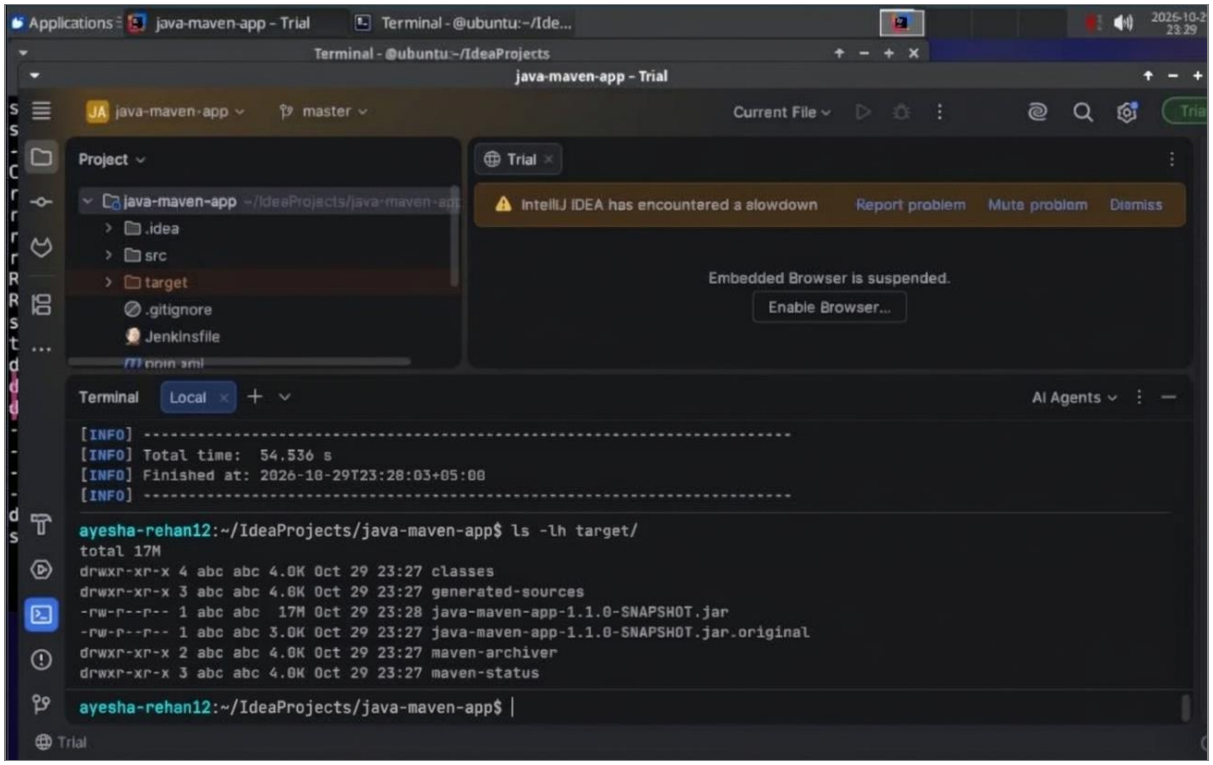


Figure: target directory showing java-maven-app-1.1.0-SNAPSHOT.jar.

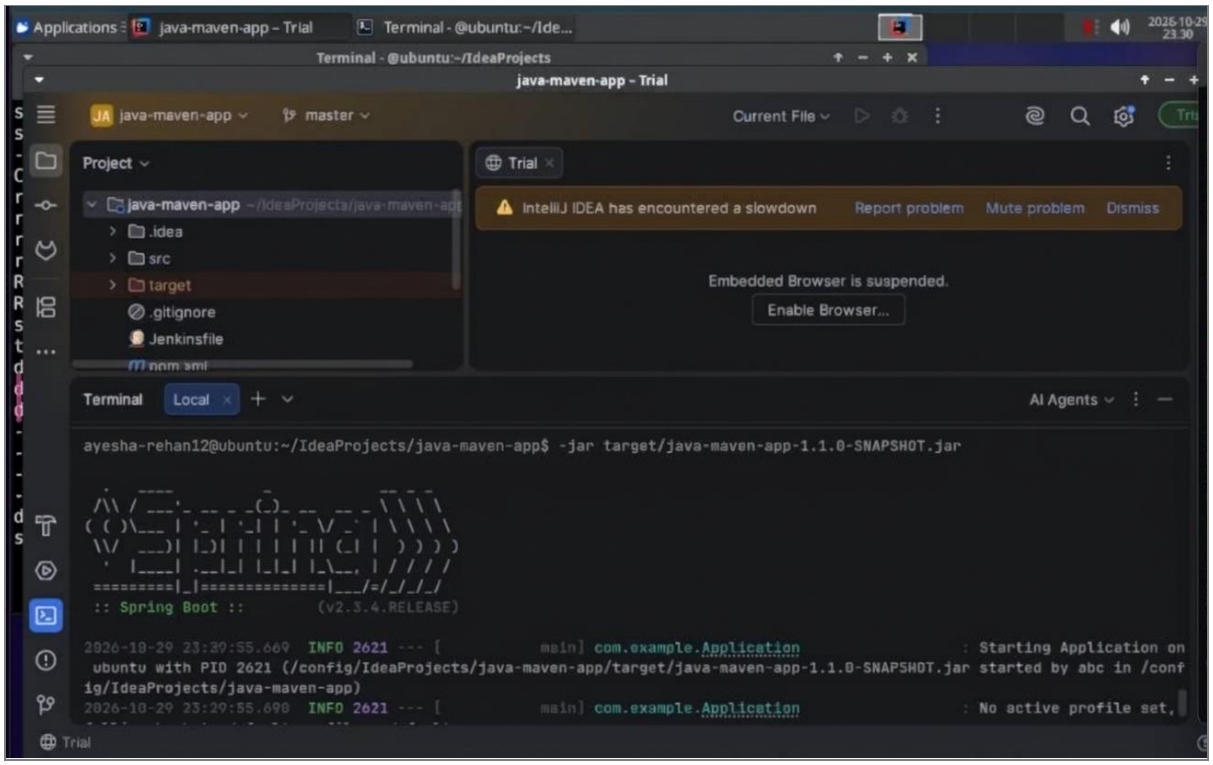


Figure: JAR started with java -jar, showing successful application startup.



Figure: Application stopped with Ctrl+C and prompt returned.

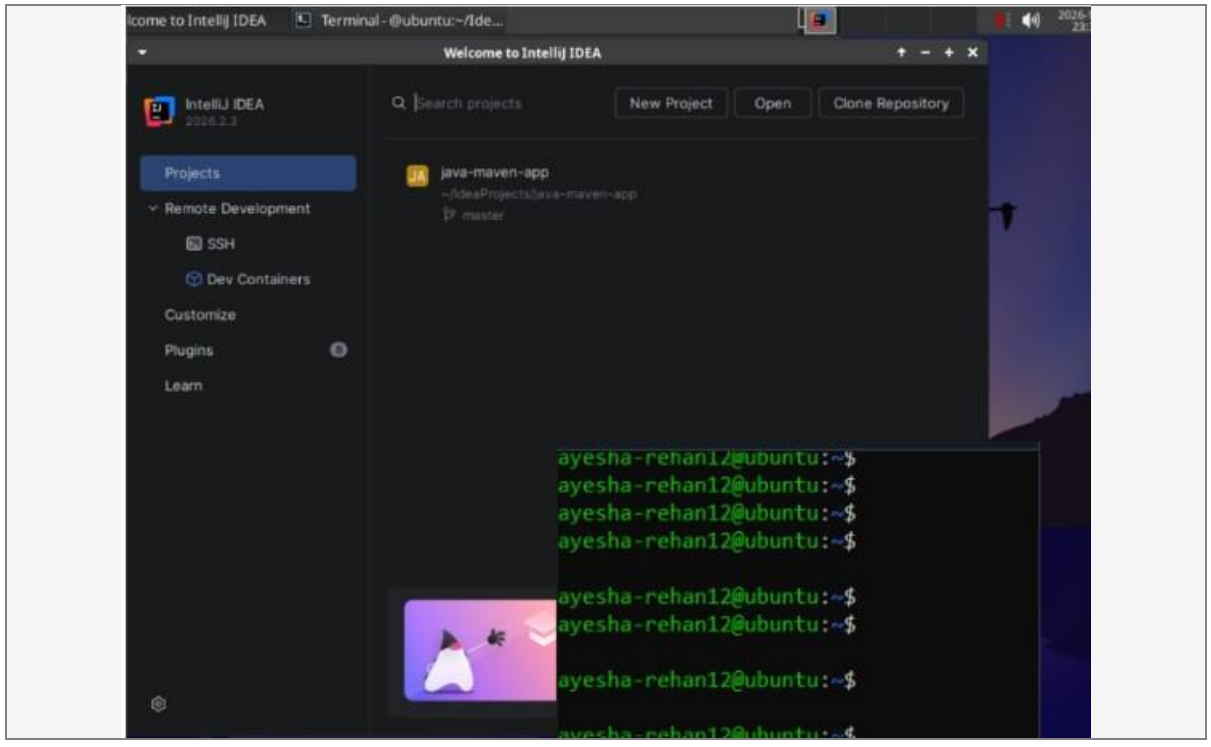


Figure: Maven project closed, showing the IntelliJ Welcome screen.

Task 3: Build and Run the Gradle Project

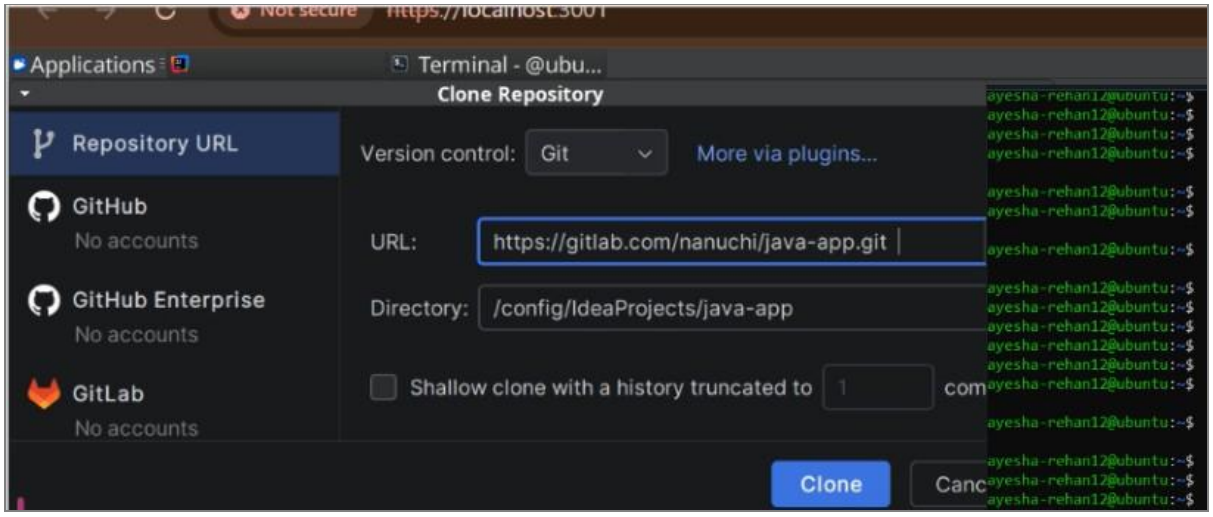


Figure: Clone dialog showing the java-app URL and destination directory.

```
ayesha-rehan12@ubuntu:~/IdeaProjects/java-app$ ./gradlew build _

> Task :compileJava
warning: [options] source value 8 is obsolete and will be removed in a future release
warning: [options] target value 8 is obsolete and will be removed in a future release
warning: [options] To suppress warnings about obsolete options, use -XLint:-options.
3 warnings

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.5/userguide/command_line_interface#sec:command_line_warnings in the Gradle
documentation.

BUILD SUCCESSFUL in 17m 49s
5 actionable tasks: 5 executed
ayesha-rehan12@ubuntu:~/IdeaProjects/java-app$ _
```

Figure: ./gradlew build completed with BUILD SUCCESSFUL on JDK 16.

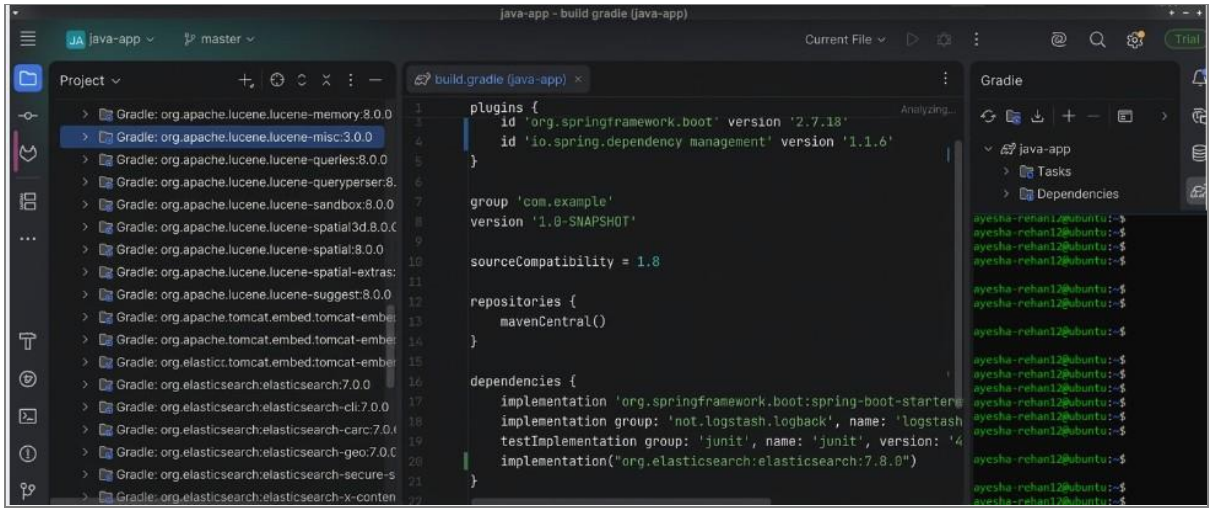


Figure: Elasticsearch 7.0.0 added in build.gradle and synchronized.

```
Terminal   Local                                                    AI Agents
ayesha-rehan12@ubuntu:~/IdeaProjects/java-app$  ./gradlew build
Starting a Gradle Daemon, 11 busy and 2 incompatible and 1 stopped Daemons could not be reused, use --status for details

> Task :compileJava
warning: [options] source value 8 is obsolete and will be removed in a future release
warning: [options] target value 8 is obsolete and will be removed in a future release
warning: [options] To suppress warnings about chsolete options, use -Xlint:-options.
3 warnings

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.5/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documer
tation.

BUILD SUCCESSFUL in 11m 43s
5 actionable tasks: 3 executed, 2 up-to-date
ayesha-rehan12@ubuntu:~/IdeaProjects/java-app$
ayesha-rehan12@ubuntu:~/IdeaProjects/java-app$
ayesha-rehan12@ubuntu:~/IdeaProjects/java-app$
```

Figure: ./gradlew build re-run after the dependency change, with BUILD SUCCESSFUL.

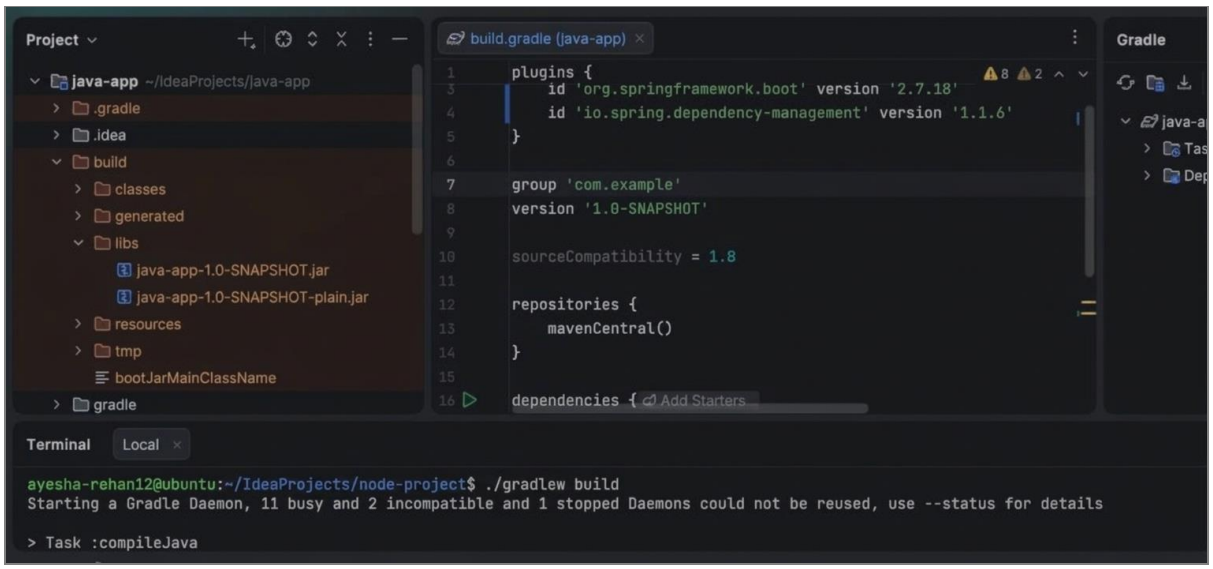


Figure: build/libs directory showing java-app-1.0-SNAPSHOT.jar.

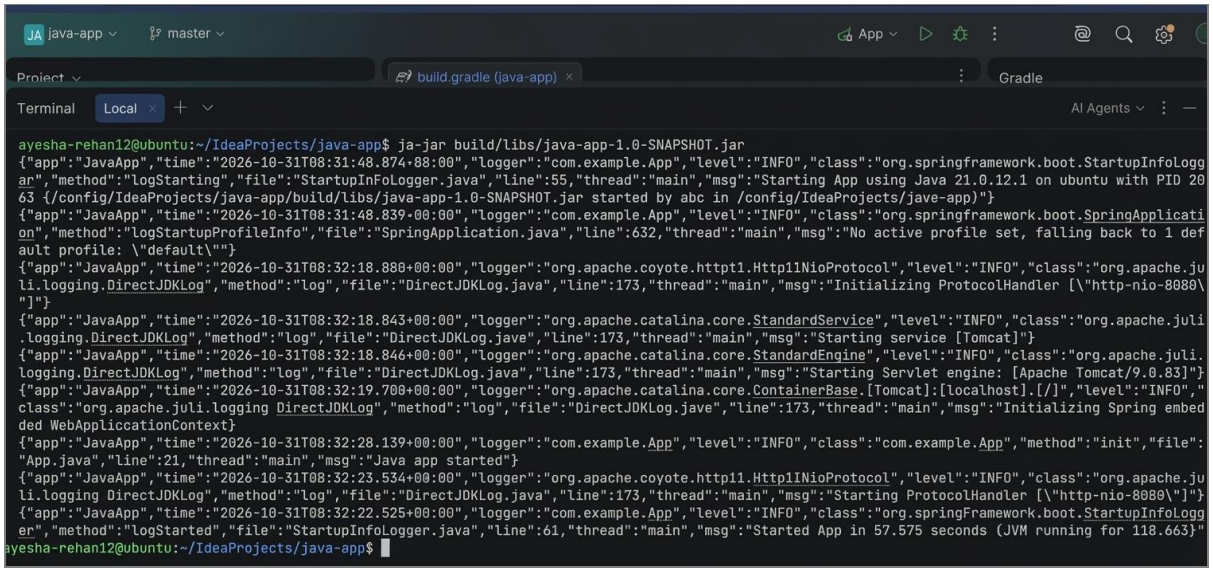


Figure: JAR started with java -jar, showing application startup.

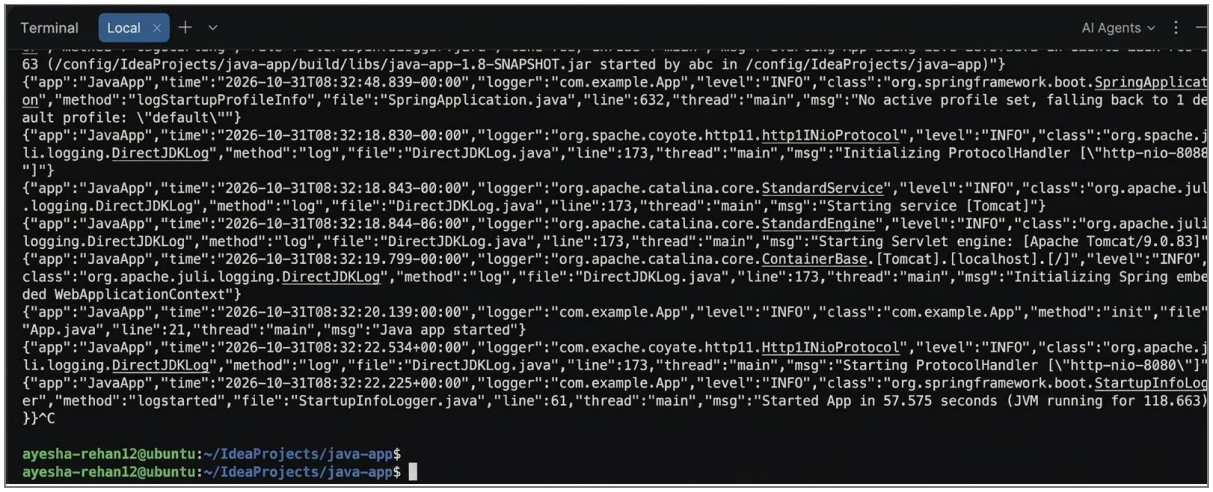


Figure: Application stopped with Ctrl+C and prompt returned.



Figure: Gradle project closed, showing the Welcome screen.

Task 4: Install and Run the Node.js Project

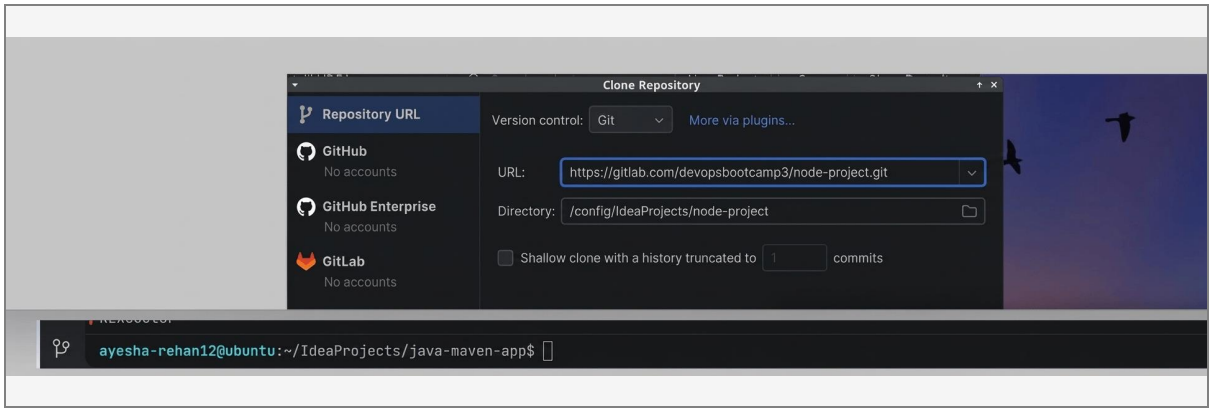


Figure: Clone dialog showing the node-project URL and destination directory.

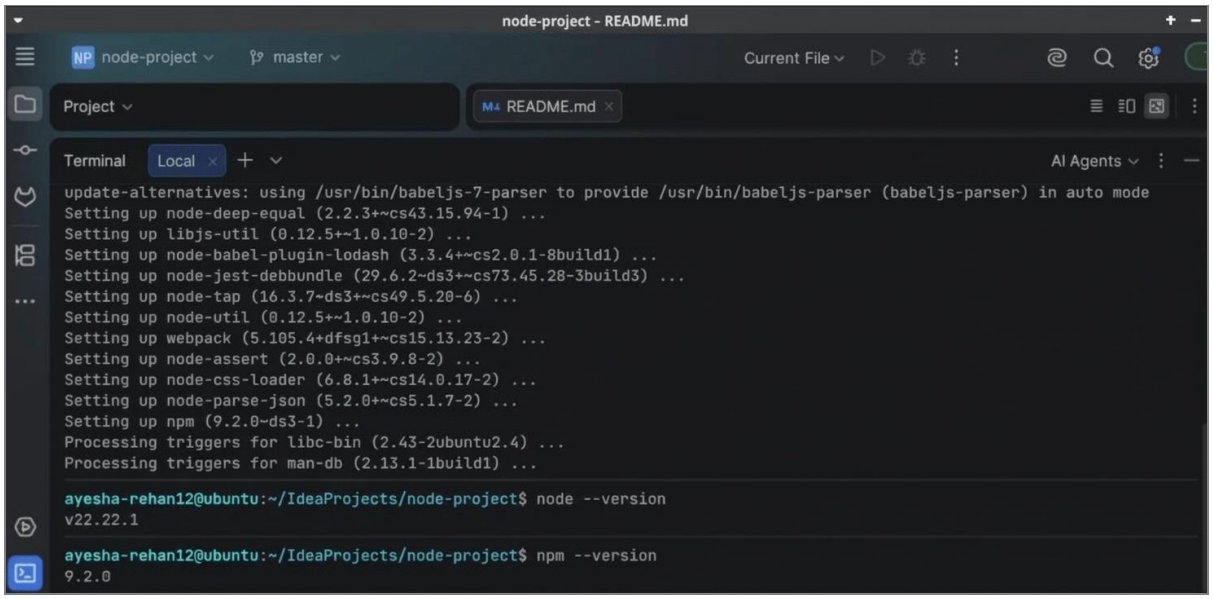


Figure: Node.js and npm installed, with versions shown.

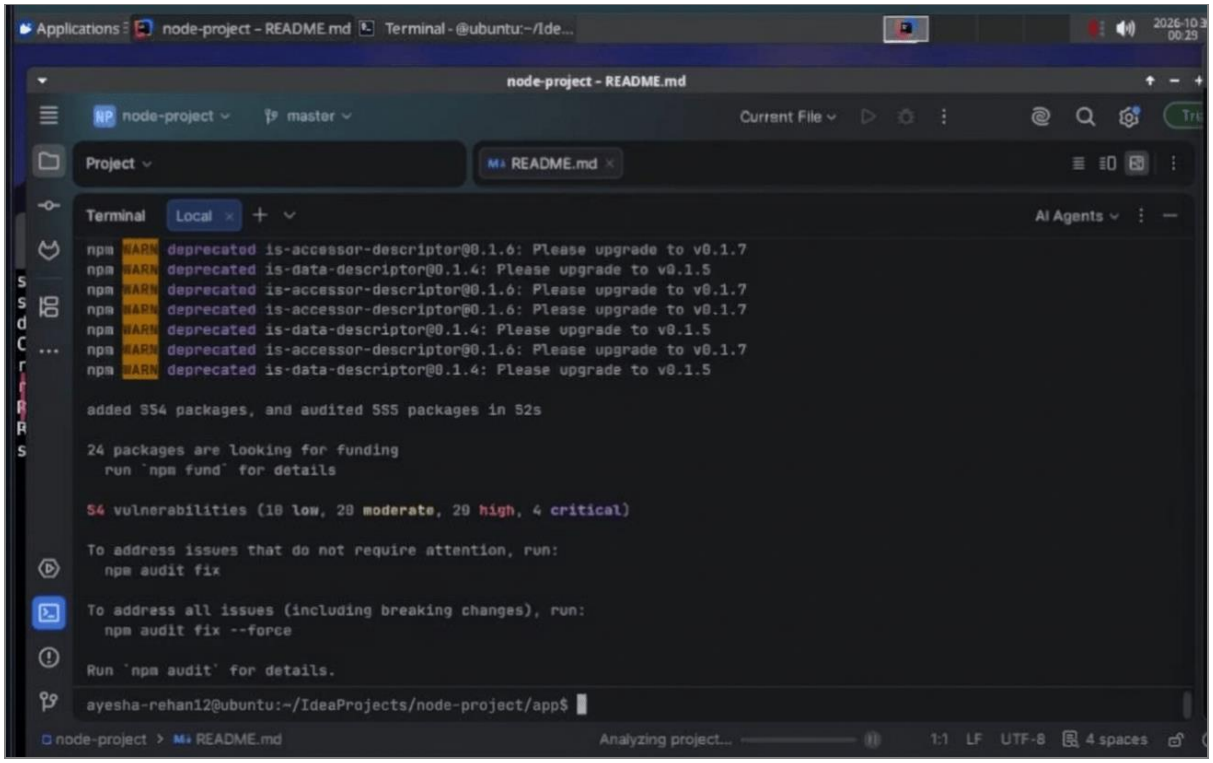


Figure: npm install completed in the app directory.



Figure: npm start showing the app listening on port 4000.

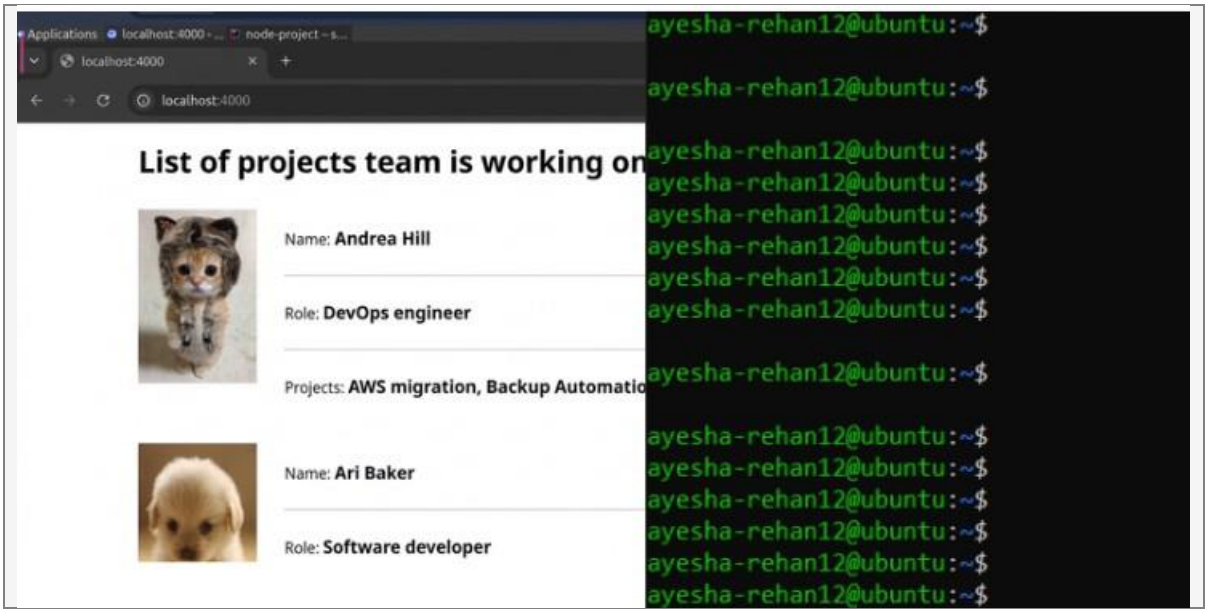


Figure: Node.js project page at http://localhost:4000 in the container browser, with IntelliJ terminal visible.

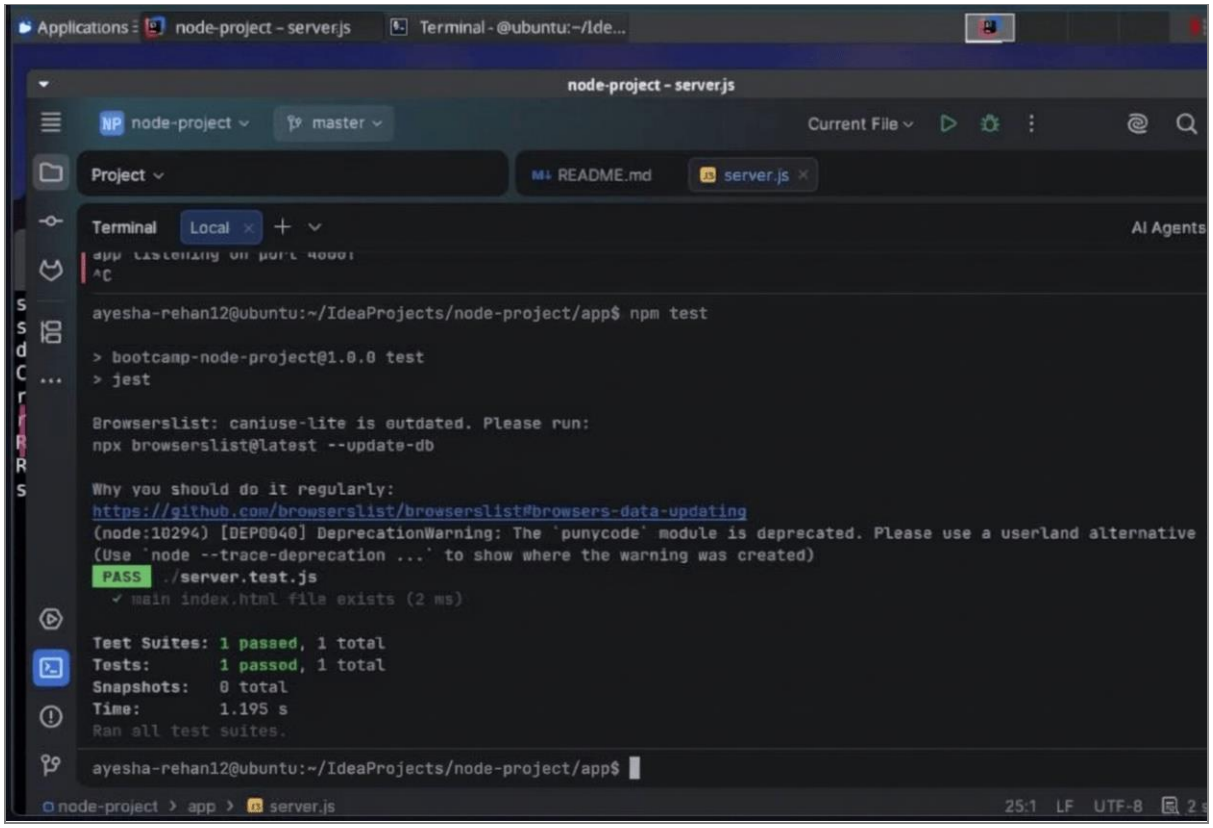


Figure: npm test passing.

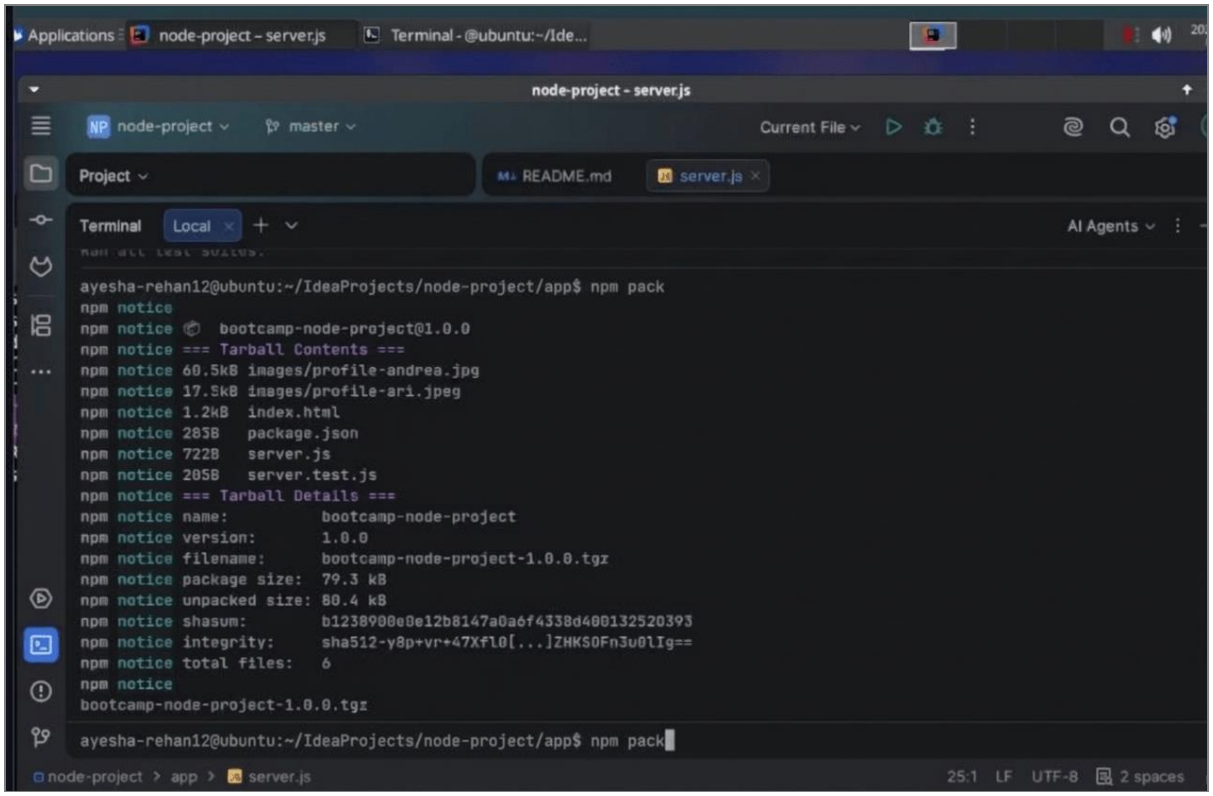


Figure: npm pack generating bootcamp-node-project-1.0.0.tgz.

Task 5: Build the React and Node.js API Artifact

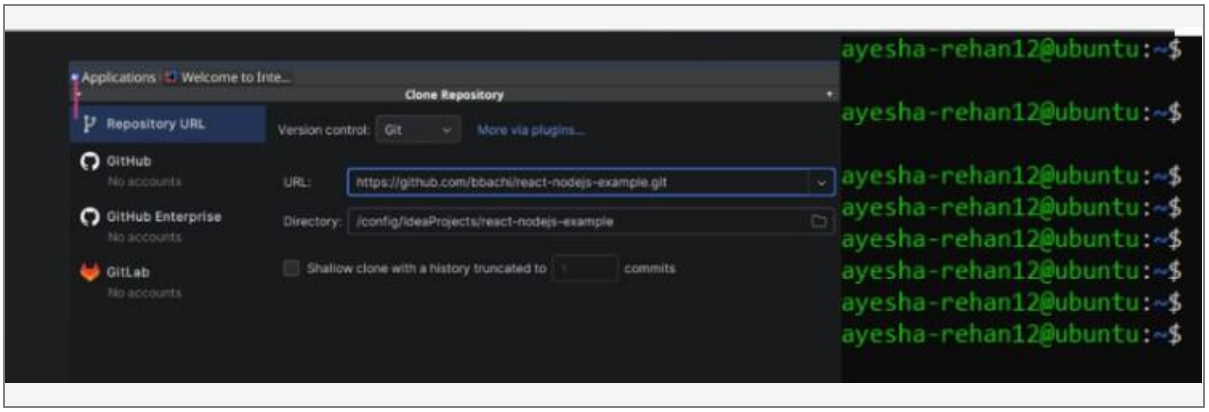


Figure: Clone dialog showing the react-nodejs-example URL and destination directory.

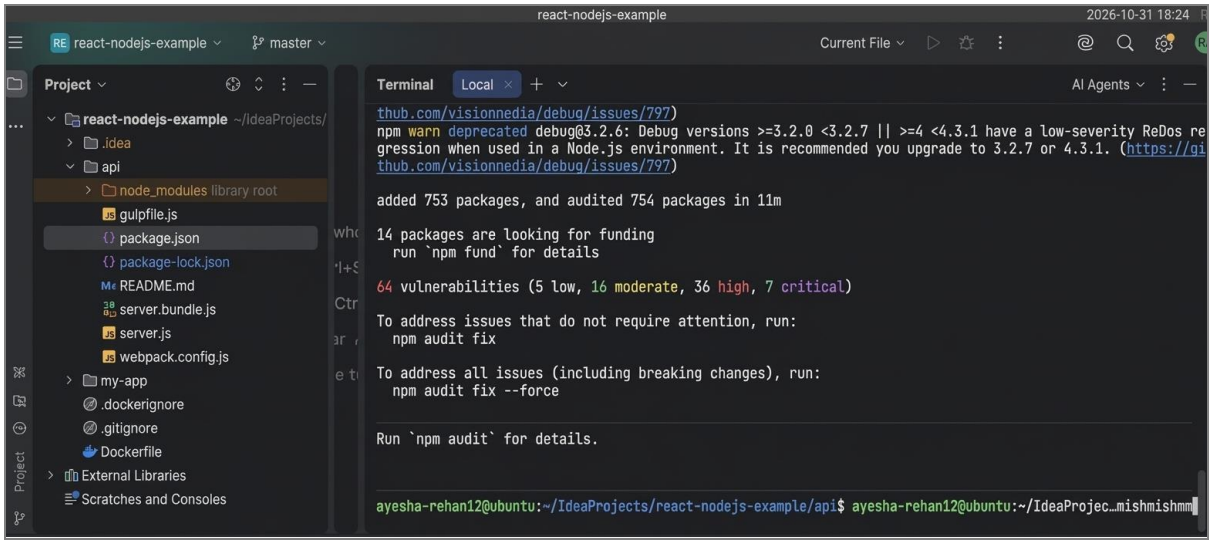


Figure: npm install completed in api, with node_modules present.

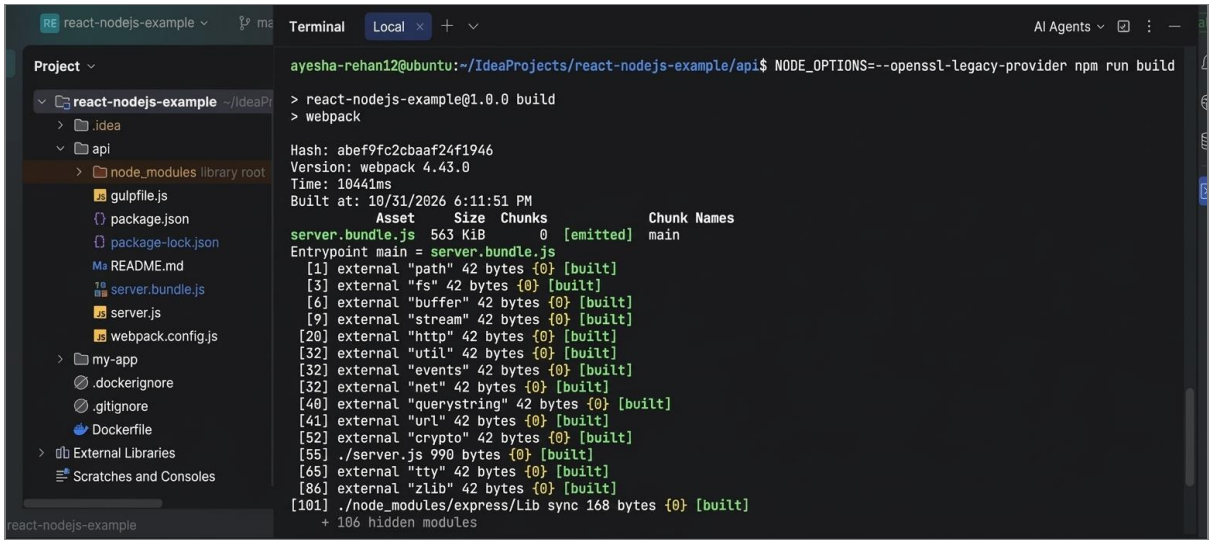


Figure: npm run build succeeded, with server.bundle.js in the api directory.

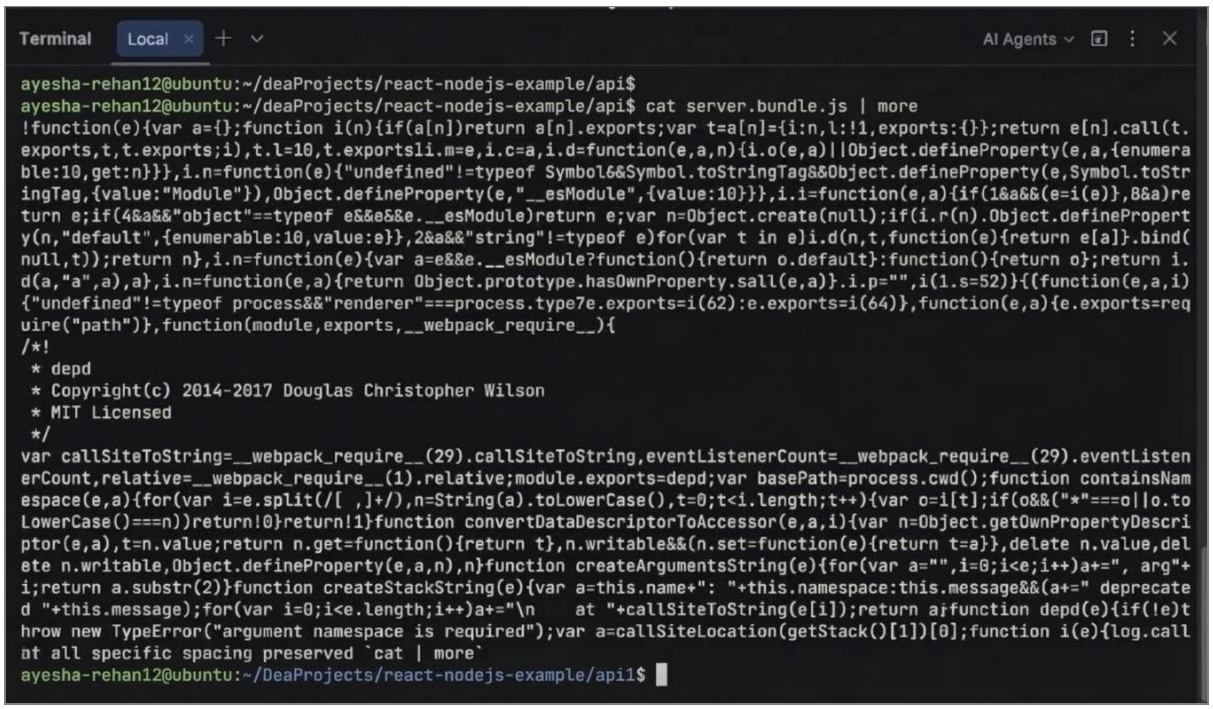


Figure: Bundled JavaScript viewed with cat server.bundle.js | more.

Task 6: Build a Docker Image for the Node.js Project



Figure: docker --version showing the installed Docker version.

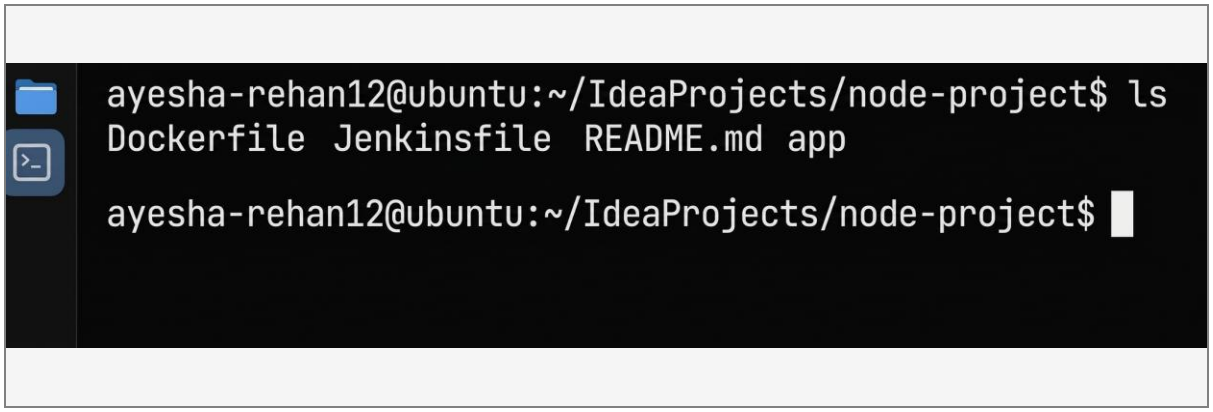


Figure: ls output showing the Dockerfile in the repository root.

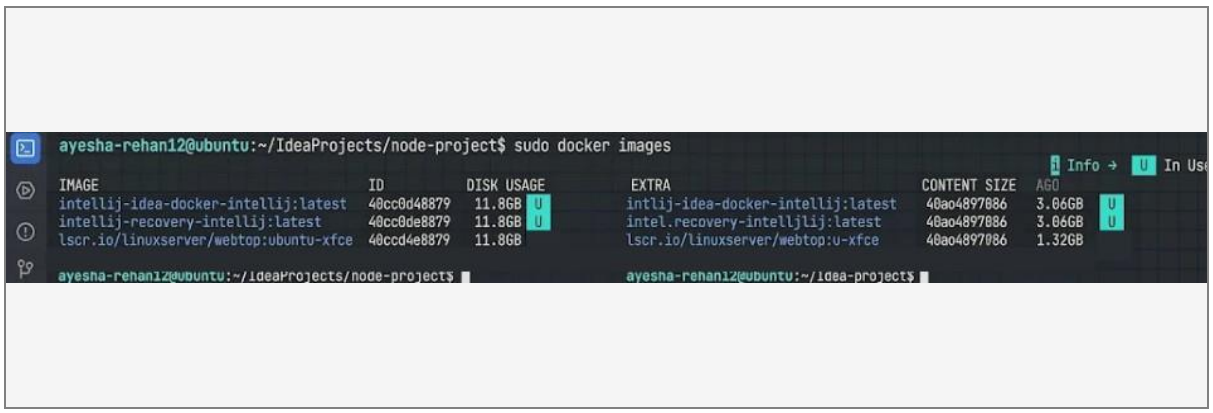


Figure: Docker image list before building.

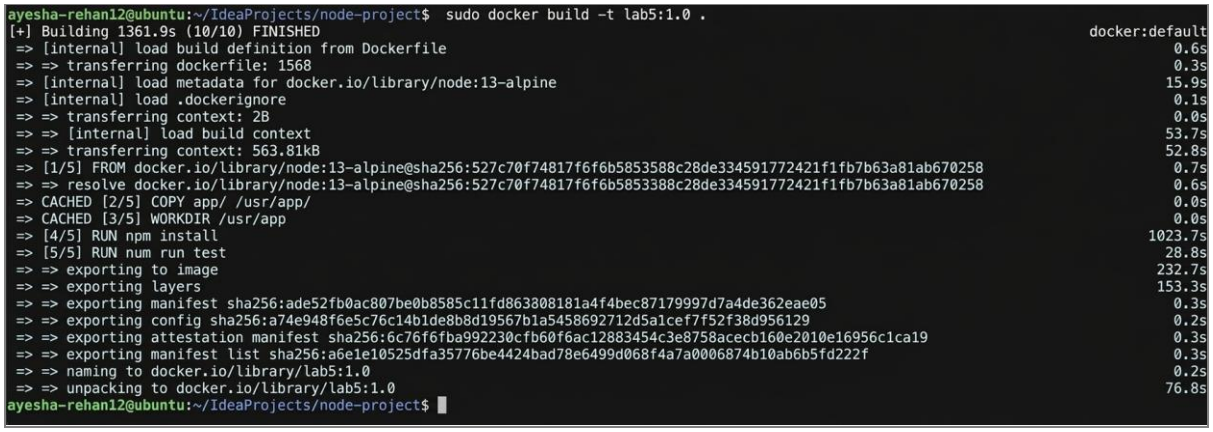


Figure: sudo docker build -t lab5:1.0 . completing successfully.

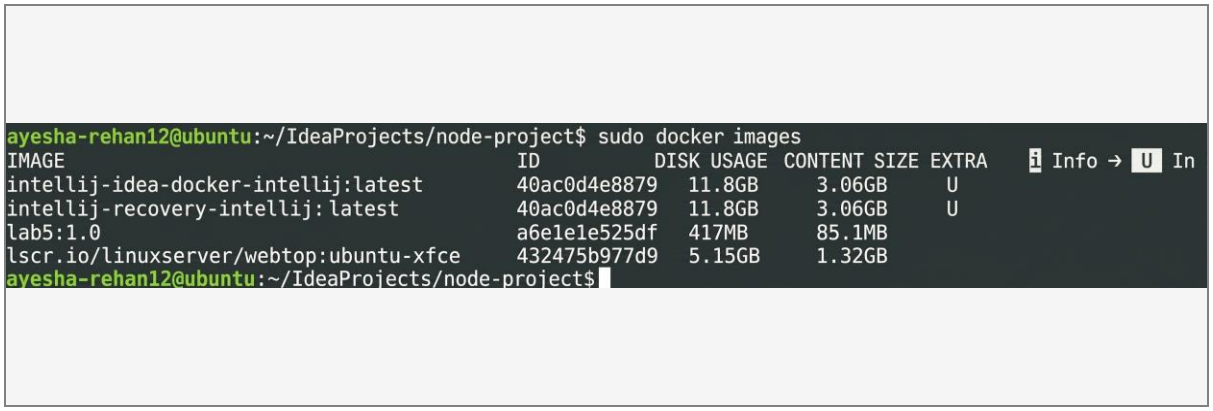


Figure: Docker image list showing repository lab5, tag 1.0, and image ID.